

T5

$$X \sim R(\theta, 2\theta)$$

a) θ MLE: $p(\theta) = \frac{1}{\theta} \mathbb{I}_{\{\theta \leq x \leq 2\theta\}}$

$$L(\theta) = \prod_{i=1}^n p(\theta) = \frac{1}{\theta^n} \mathbb{I}_{\{\theta \leq x_i \leq 2\theta\}}$$

$$\sup_{\theta} L(\theta) \rightarrow \hat{\theta}_1 = \max_{\theta} x_i = \frac{x_{\max}}{2} \quad \theta \geq \frac{x_{\max}}{2}$$

MLE

$$\hat{\theta}_1(\theta) = \hat{\theta}_1 = \bar{x}$$

$$\begin{aligned} \hat{\theta}_1 = MLE &= \int_{-\infty}^{\infty} x \cdot \frac{1}{\theta} dx = \frac{x^2}{2} \cdot \frac{1}{\theta} \Big|_{\theta}^{2\theta} = \frac{4\theta^2}{2\theta} - \frac{\theta^2}{2\theta} = \\ &= \frac{3\theta}{2} \end{aligned}$$

$$\frac{3\theta}{2} = \bar{x}$$

$$\hat{\theta}_1 = \frac{2}{3} \bar{x}$$

b) $M(\hat{\theta}_1) = \frac{1}{2} \int_{-\infty}^{\infty} x dF = \frac{1}{2} \int_{\theta}^{2\theta} x \cdot n F^{n-1}(\frac{x}{\theta}) \cdot \frac{1}{\theta} dx =$

$$= \frac{1}{2} \int_{\theta}^{2\theta} x \cdot n \cdot \left(\frac{x-\theta}{\theta} \right)^{n-1} \frac{1}{\theta} dx =$$

$$\frac{x-\theta}{\theta} = t$$

$$x = \theta t + \theta = \theta(1+t)$$

$$dx = \theta dt$$

$$= \frac{1}{2} \int_{\theta}^{2\theta} n \theta(1+t) \cdot t^{n-1} \frac{\theta}{\theta} dt = \frac{1}{2} \theta n \int_0^1 (1+t) t^{n-1} dt =$$

$$= \frac{1}{2} \theta n \int_0^1 t^{n-1} dt + \frac{1}{2} \theta n \int_0^1 t^n dt = \frac{1}{2} \theta n \left(\frac{t^n}{n} + \frac{t^{n+1}}{n+1} \right) \Big|_0^1 =$$

$$M\tilde{\theta}_1 = \theta_n \cdot \frac{n+1+n}{2(n+1)} = \theta \cdot \frac{2(n+1)}{2(n+1)} \sim \text{сильно}$$

$$\tilde{\theta}_1 = \frac{2(n+1)}{2n+1} \tilde{\theta}_1 - \text{не очень хорошая}$$

$$M\tilde{\theta}_2 = M \frac{2}{3} \bar{x} = \frac{2}{3} M \bar{x} = \frac{2}{3} M \bar{\varepsilon} = \frac{2}{3} \cdot \frac{3}{2} \theta = \theta$$

$$D\tilde{\theta}_2 = D \frac{2}{3} \bar{x} = \frac{4}{9} D \bar{x} =$$

$$= \frac{4}{9} D \left(\frac{\varepsilon}{n} \right) = \frac{4}{9n} D \varepsilon = \frac{4}{9n} \frac{(2\theta - \theta)^2}{12} =$$

$$= \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \quad (1)$$

$\tilde{\theta}_2$ - не смещ. (2)

$\Rightarrow \tilde{\theta}_2$ - хорошая оценка по дост. усл.

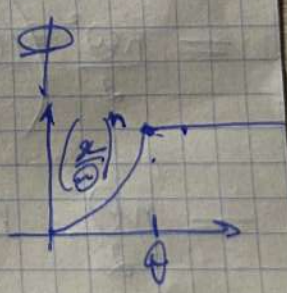
$$\forall \varepsilon > 0 \forall \theta \in \mathbb{R} \quad \lim_{n \rightarrow \infty} P\left(\left|\frac{x_{\max}}{2} - \theta\right| \geq \varepsilon\right) = 0$$

$$P\left(\left|\frac{x_{\max}}{2} - \theta\right| \geq \varepsilon\right) = P\left(\frac{x_{\max}}{2} \geq \theta + \varepsilon\right) + P\left(\frac{x_{\max}}{2} \leq \theta - \varepsilon\right)$$

$$= P\left(\frac{\theta - \varepsilon}{2}\right) = \begin{cases} \frac{\theta - \varepsilon}{2} \leq 0 : 0 \\ 0 \leq \frac{\theta - \varepsilon}{2} \leq \theta : \left(\frac{\theta - \varepsilon}{2\theta}\right)^n \end{cases}$$

$$\xrightarrow{n \rightarrow \infty} 0$$

$\tilde{\theta}_2$ - хорошая



$$c) \mathcal{M}_1^{\tilde{\alpha}} = \frac{1}{2} \int_{-\infty}^{\infty} x^2 dx = \frac{1}{2} \int_{-\infty}^{\infty} x^2 \left(\frac{x-\theta}{\theta} \right)^{n-1} dx \quad \text{---}$$

$$\frac{x-\theta}{\theta} = t$$

$$x = \theta t + \theta = \theta(1+t)$$

$$dx = \theta dt$$

$$\text{---} \Rightarrow \frac{1}{2} \int_0^1 \theta^2 (1+t)^2 \cdot t^{n-1} \cdot \frac{1}{\theta} \cdot \theta dt =$$

$$= \frac{n\theta^2}{2} \int_0^1 (1+t)^2 \cdot t^{n-1} dt =$$

$$= \frac{n\theta^2}{2} \left(\int_0^1 t^{n-1} dt + 2 \int_0^1 t^n dt + \int_0^1 t^{n+1} dt \right) =$$

$$= \frac{n\theta^2}{2} \left(\left. \frac{t^n}{n} \right|_0^1 + 2 \cdot \left. \frac{t^{n+1}}{n+1} \right|_0^1 + \left. \frac{t^{n+2}}{n+2} \right|_0^1 \right) =$$

$$= \frac{n\theta^2}{2} \left(\frac{1}{n} + \frac{2}{n+1} + \frac{1}{n+2} \right) =$$

$$= \frac{n\theta^2}{2} \left(\frac{(n+1)(n+2) + 2n(n+2) + n(n+1)}{n(n+1)(n+2)} \right) =$$

$$= \frac{n\theta^2}{2} \cdot \frac{n^2 + 3n + 2 + 2n^2 + 4n + n^2 + n}{n(n+1)(n+2)} =$$

$$= \frac{n\theta^2}{2} \cdot \frac{4n^2 + 8n + 2}{n(n+1)(n+2)} = \theta^2 \cdot \frac{2n^2 + 4n + 1}{(n+1)(n+2)}$$

$$\mathcal{D}_{\tilde{\alpha}_1}^{\tilde{\alpha}} = \theta^2 \left(\frac{2n^2 + 4n + 1}{(n+1)(n+2)} - \left(\frac{2n+1}{(n+1)} \right)^2 \right) =$$

$$\Theta^2 \left(\frac{2(2n^2 + 4n + 1)(n+1) - (2n+1)^2(n+2)}{4(n+1)^2(n+2)} \right) =$$

$$= \Theta^2 \cdot \frac{(4n^3 + 8n^2 + 2n + 4n^2 + 8n + 2) - (4n^3 + 4n^2 + 8n + 2)}{4(n+1)^2(n+2)} =$$

$$= \Theta^2 \cdot \frac{(4n^3 + 12n^2 + 10n + 2) - (4n^3 + 4n^2 + 8n + 2)}{4(n+1)^2(n+2)} =$$

$$= \frac{\Theta^2 n}{4(n+1)^2(n+2)}$$

$$\tilde{D}\tilde{\Theta}_2' = \left(\frac{2(n+1)}{2n+1} \right)^2 \cdot \frac{\Theta^2 n}{4(n+1)^2(n+2)}$$

$$= \frac{4(n+1)^2 \Theta^2 - n}{4(n+1)^2(n+2)(2n+1)^2} = \frac{n \cdot \Theta^2}{(n+2)(2n+1)^2}$$

$$\underline{I(\Theta) = \int_{-\infty}^{\infty} \frac{1}{\Theta^2} \cdot \frac{1}{\Theta} dx = \int_{-\infty}^{\infty} \frac{1}{\Theta^3} dx = \frac{x}{\Theta^3} \Big|_{-\infty}^{\infty} =}$$

модель не регулярная $\Rightarrow$ не можем восп. кер Крамера - Рае

$$\tilde{D}\tilde{\Theta}_1' \vee \tilde{D}\tilde{\Theta}_2$$

$$\frac{n^2}{(n+2)(2n+1)^2} \not\sim \frac{\Theta^2}{d \neq n}$$

$$2 \neq n^2 \not\sim 4n^3 + \dots$$

$\tilde{D}\tilde{\Theta}_1'$ - дифференциальное,
если $\tilde{D}\tilde{\Theta}_2$
тогда
где $n \geq 3$

d)

$$y = \frac{x_{\max}}{\theta}$$

$$F_{x_{\max}} = \left(\frac{x-\theta}{\theta}\right)^n$$

$$F_y = P\left(\frac{x_{\max}}{\theta} \leq y\right) = P(x_{\max} \leq \theta y)$$

$$y \in [1; 2]$$

$$F_y = \left(\frac{\theta y - \theta}{\theta}\right)^n = (y-1)^n$$

$$F_y = \begin{cases} 0, & y < 1 \\ (y-1)^n, & 1 \leq y \leq 2 \\ 1, & y > 2 \end{cases}$$

$$p_y = n(y-1)^{n-1}, y \in [1; 2]$$

$$P(a \leq y \leq b) = \beta$$

$$\frac{x_{\max}}{b} \leq \theta \leq \frac{x_{\max}}{a}$$

гиперметрический интервал: $\left[\frac{x_{\max}}{b}, \frac{x_{\max}}{a}\right]$

$$P(y < a) = \frac{\alpha}{2}$$

$$P(y > b) = \frac{\alpha}{2}$$

$$(a-1)^n = \frac{\alpha}{2}$$

$$(b-1)^n = 1 - \frac{\alpha}{2}$$

$$a = 1 + \left(\frac{\alpha}{2}\right)^{1/n}$$

$$b = 1 + \left(1 - \frac{\alpha}{2}\right)^{1/n}$$

гиперметрический интервал: $\left[\frac{x_{\max}}{1 + \left(1 - \frac{\alpha}{2}\right)^{1/n}}, \frac{x_{\max}}{1 + \left(\frac{\alpha}{2}\right)^{1/n}}\right]$

e)

$$\mu_{\xi} = \frac{\theta + 2\theta}{2} = \frac{3\theta}{2}$$

$$\sigma^2 = \frac{(2\theta - \theta)^2}{12}$$

DMU

$$\sqrt{n}(\bar{X} - \mu_{\xi}) \sim N(0, \frac{\theta^2}{12})$$

$$\tilde{\theta} = \frac{2}{3}\bar{X} \Rightarrow \bar{X} = \frac{3}{2}\tilde{\theta}$$

$$\sqrt{n}\left(\frac{3}{2}\tilde{\theta} - \frac{3}{2}\theta\right) \sim N(0, \frac{\theta^2}{12})$$

$$\sqrt{n}(\tilde{\theta} - \theta) \sim N(0, \frac{\theta^2}{27})$$

$$\frac{\sqrt{n}(\tilde{\theta} - \theta)}{\tilde{\theta}/\sqrt{27}} \sim N(0, 1)$$

$$P\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\tilde{\theta} - \theta}{\tilde{\theta}/\sqrt{27}} \sqrt{n} \leq z_{1+\frac{\alpha}{2}}\right) = \beta$$

$$\theta \in \left[\tilde{\theta} - z_{1-\frac{\alpha}{2}} \cdot \frac{\tilde{\theta}}{\sqrt{27n}}; \tilde{\theta} + z_{1+\frac{\alpha}{2}} \cdot \frac{\tilde{\theta}}{\sqrt{27n}}\right]$$

$$\beta = 0,95$$

$$z_{0,975} \approx 1,96$$

$$\theta \in \left(1 - \frac{1,96}{\sqrt{27n}}; 1 + \frac{1,96}{\sqrt{27n}}\right)$$

T6

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta}; & x \geq 1, \theta > 1 \\ 0, & x < 1 \end{cases}$$

$$a) L(\theta) = \prod_{i=1}^n \frac{\theta-1}{x_i^\theta} = (\theta-1)^n \cdot \prod_{i=1}^n x_i^{-\theta}; x_i \geq 1$$

$$\ln L(\theta) = n \ln(\theta-1) - \theta \sum_{i=1}^n \ln x_i$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta-1} - \sum_{i=1}^n \ln x_i$$

$$\frac{n}{\theta-1} = \sum_{i=1}^n \ln x_i$$

$$\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} < 0 \text{ for } \theta > 1$$

$$\theta-1 = \frac{n}{\sum_{i=1}^n \ln x_i}$$

$$\Rightarrow \text{max}$$

$$\hat{\theta} = \frac{n}{\sum \ln x_i} + 1$$

$$c) \int_{-\infty}^t \frac{\theta-1}{x^\theta} dx = \frac{1}{1-\theta} \cdot \frac{\theta-1}{x^{\theta-1}} \Big|_{-\infty}^t = -\frac{1}{\theta-1} \Big|_{-\infty}^t =$$

$$= 0 - \frac{1}{t^{\theta-1}} + 1 = 1 - t^{1-\theta} = F(t)$$

$$X_{\frac{1}{2}} \Rightarrow F(X_{\frac{1}{2}}) = 1 - X_{\frac{1}{2}}^{1-\theta} = 0.5$$

$$X_{\frac{1}{2}}^{1-\theta} = \frac{1}{2}$$

$$X_{\frac{1}{2}} = \sqrt[1-\theta]{\frac{1}{2}}$$

$$\sqrt{n} (g(\hat{\theta}) - g(\theta)) \sim N(0, \nabla^T g(\theta) I^{-1}(\theta) \nabla g(\theta))$$

$$I(\theta) = -\mathcal{M} \left[\frac{\partial^2 \ln p}{\partial \theta^2} \right] = -\mathcal{M} \left[-\frac{1}{(\theta-1)^2} \right] = \frac{1}{(\theta-1)^2}$$

$$I(\theta) = \mathcal{M} \left(\frac{\partial \ln p}{\partial \theta} \right)^2 = \mathcal{M} \left[\frac{1}{(\theta-1)^2} \right] - \frac{2}{\theta-1} \mathcal{M} \ln x + \mathcal{M} [(\ln x)^2] =$$

$$= \frac{1}{(1-\theta)^2} - \frac{2}{\theta-1} \cdot \frac{1}{\theta-1} + \frac{2}{(\theta-1)^2} = \frac{1}{(1-\theta)^2}$$

$$\sqrt{n} \left(\frac{n}{\sum \ln x_i} + 1 - \theta \right) \sim N(0, \frac{(1-\theta)^2}{\mathcal{O}^2(\theta)})$$

$$\frac{\sqrt{n} \left(\frac{n}{\sum \ln x_i} + 1 - \theta \right)}{1 - \theta} \sim N(0, 1)$$

$$P \left(-z_{1-\frac{\alpha}{2}} \leq \frac{\hat{\theta} - \theta}{\sqrt{(1-\theta)^2/n}} \leq z_{1-\frac{\alpha}{2}} \right) = \beta$$

$$\tilde{\Theta} - z_{1-\frac{\alpha}{2}} \cdot \frac{\tilde{\Theta}-1}{\sqrt{n}} \leq \Theta \leq \tilde{\Theta} + z_{1-\frac{\alpha}{2}} \cdot \frac{\tilde{\Theta}-1}{\sqrt{n}}$$

~~$$\tilde{\Theta} \in [-1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}, 1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}]$$~~

$$\Theta \in [\tilde{\Theta} - 1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}; \tilde{\Theta} + 1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}]$$

$I(\Theta)$ — непрерывная
 $I(\Theta) > 0$ — регулярная модель

$$\frac{\partial}{\partial \Theta} \int F dx = \int \frac{\partial}{\partial \Theta} F dx$$

т.к. интегралы св. равномерно на компакте Θ $\square$

$$P(\Theta_L \leq \Theta \leq \Theta_R) = \beta$$

$$P(2^{1/\Theta_R-1} \leq m \leq 2^{1/\Theta_L-1}) = \beta$$

$$\Theta_R = \tilde{\Theta} + 1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}$$

$$\Theta_L = \tilde{\Theta} - 1.96 \frac{\tilde{\Theta}-1}{\sqrt{n}}$$